Glink

TECHNICAL PROPOSAL · OCTOBER 2026

Building Glink for *production.*

Architecture, the stack and why, payments and automated escrow, video storage, security, running costs and a 16-week plan to take the working platform from pilot to scale.

1 Executive summary
2 Architecture: today and target
3 Stack and tools
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7 Video storage and delivery
8 Payments: money in, money out
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Prepared for
Glink founders and technical reviewers

Status
Proposal for approval

Codebase
github.com/philippython/glink

1 · EXECUTIVE SUMMARY

A working platform, ready to be hardened and scaled.

Glink is a two-sided UGC marketplace: brands write one brief, Glink matches it to the creators most likely to deliver, checks every video against the brief before a brand sees it, and pays creators from escrow on approval — with automatic bonuses when an ad wins. The core product is built and tested. This proposal sets out how we take it to production and then to scale, with particular attention to **payments** and **video storage**.

Automated tests passing
36
auth, matching, payments, escrow automation

Match factors, explained
8
learned from real ad results

Plan to production scale
16 wks
five overlapping phases

Pilot hosting cost
~$10/mo
one server + disk; rises with use

What we propose

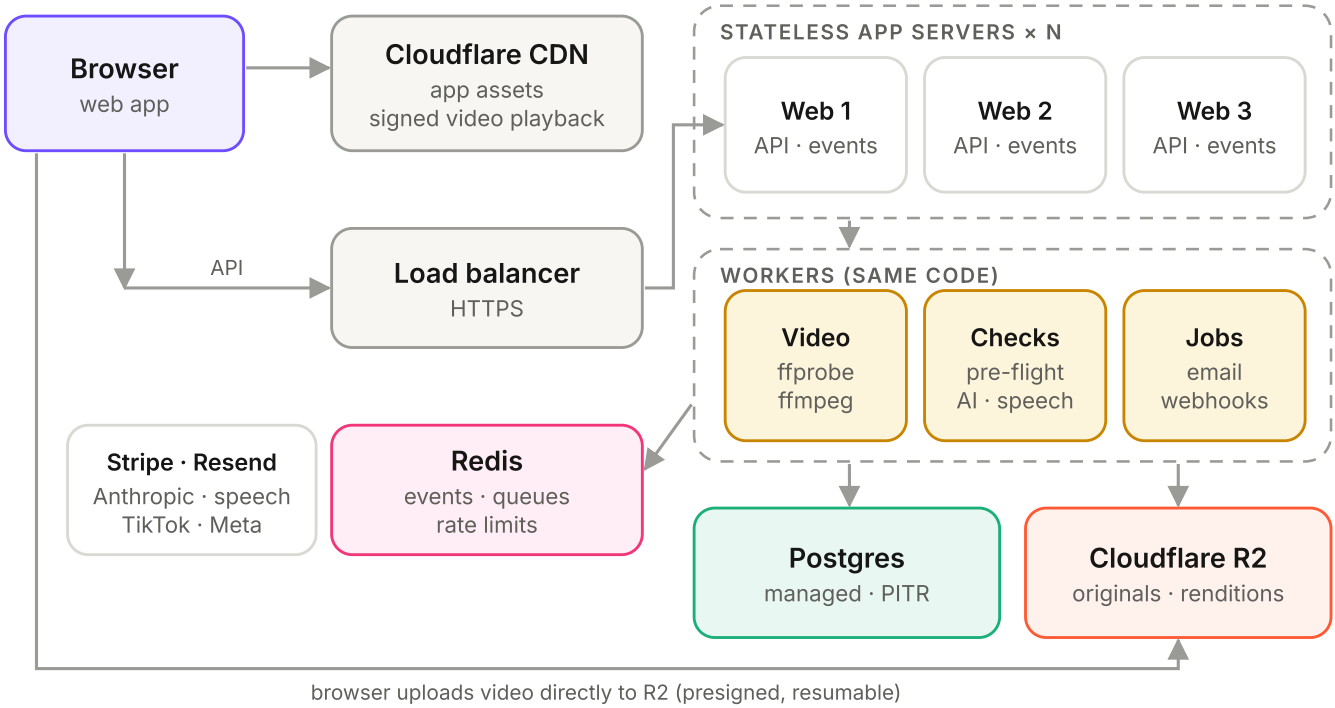
AREA	TODAY	PROPOSED
Payments	Stripe Checkout top-ups and Connect payouts; automated escrow (reminders, auto-approval, clearance, auto-payouts); idempotent webhooks	Go live on Stripe; add platform fee, refunds, dispute handling, nightly reconciliation, and a provider adapter so Paystack can serve African markets
Video storage	Files on the server's persistent disk; uploads pass through the server	Direct-to-storage resumable uploads to Cloudflare R2; server-side verification and transcoding; signed CDN playback; tiered retention
Data & scale	SQLite on one server; live updates in memory	Managed Postgres with point-in-time recovery, Redis for events and job queues, stateless app servers behind a load balancer
Quality checks	Format and script checks; optional Claude review of sampled frames	Add speech-to-text so spoken lines are checked without a script; move checks into background workers
Integrations	Ad results entered by hand; follower counts self-reported	Import results from TikTok and Meta ad accounts; verify creator channels via platform sign-in
Operations	Deployable to Render with guides and a Blueprint	CI/CD, error tracking, uptime monitoring, backups with tested restores, staging environment, security review

Recommendation. Launch the pilot now on the current single-server architecture with Stripe and email switched on (Phase 1, three weeks). Start the video pipeline in parallel, because it removes the main scaling and memory constraint. Move to Postgres and multiple servers when usage justifies it — the code is structured so this is a contained change, not a rewrite.

2 · ARCHITECTURE: TODAY AND TARGET

Stateless servers, managed data, video off the request path.

The target keeps the same code and API, but moves state out of the server process. Large files never pass through the app servers, slow work runs in background workers, and any app server can answer any request — so we can add servers as traffic grows.



CHANGE	CHOICE	REASON
Database	Managed Postgres (e.g. Neon, Render or Supabase) with point-in-time recovery	Multiple servers, concurrent writes, backups to the minute. The SQL is close to SQLite's, so the move is a contained migration
Object storage	Cloudflare R2 (S3-compatible)	No egress fees — the dominant cost of video at scale; S3 API keeps us portable to AWS S3
CDN	Cloudflare in front of the app and R2	Fast global delivery of the app and video; signed URLs keep videos private
Events & jobs	Redis: pub/sub for live updates, BullMQ queues, shared rate limits	Live updates reach users on any server; retries and back-off for every background job
Workers	Same codebase, separate process: transcode, pre-flight, email, webhooks, reconciliation, the payout sweep	Keeps API latency low and isolates heavy CPU work
Hosting	Render (or Fly/Railway) web + worker services; optional AWS later	Low operational load now; containers move anywhere

Today: one Node service serves the app and API, with SQLite and videos on a persistent disk — ideal for a pilot, but limited to one server, uploads buffered in memory, and data that only survives if a disk is attached (on Render's free plan it is wiped on every restart). The target removes each limit. Serverless hosts (Netlify, Vercel) suit only isolated pieces, not the core.

3 · STACK AND TOOLS

Proven tools, chosen for a two-sided video marketplace.

Each choice favours tools that are standard, well documented and portable, so Glink is never locked into one vendor. Costs are list prices at the time of writing — confirm before budgeting.

AREA	CHOICE	WHY IT'S THE BEST FIT	TRADE-OFFS	APPROX. COST	PROVEN AT
Web app	React 19 · Vite · Tailwind	+ Huge talent pool; fast builds; one design system for light and dark themes	– Client-side app needs care for SEO pages (landing can be pre-rendered)	Free (open source)	React: Meta, Airbnb, Netflix
API	Node.js 24 · Hono	+ One language end to end; Hono is tiny, fast and standards-based, runs on any host	– Smaller ecosystem than Express (but standard Fetch API)	Free	Node.js: Netflix, LinkedIn, PayPal
Database	SQLite → Postgres (Neon or Supabase)	+ SQLite: zero-ops pilot. Postgres: concurrent writes, point-in-time recovery, many servers	– Postgres move needs async data layer (Phase 3); SQLite is single-server	SQLite free · Postgres free tier, then ~$20–45/mo	Postgres: Instagram, Reddit
Auth	Built-in (scrypt, sessions, OAuth)	+ Already built and tested; no per-user fees; full control of roles and teams	– We maintain it (2FA and audit log planned)	Free	Same pattern as most Rails/Django apps
Auth (alternative)	Supabase Auth or Clerk	+ Hosted login, MFA and social sign-in out of the box	– Migration of existing users; per-user pricing at scale (Clerk); vendor lock-in	Supabase free to 50k users · Clerk free to 10k, then per user	—
Object storage	Cloudflare R2	+ S3-compatible API with **no egress fees** — the main cost of video playback	– Fewer advanced features than AWS S3	$0.015/GB-month · 10 GB free	S3 API: Netflix, Airbnb run on S3
Video processing	FFmpeg workers	+ Industry-standard, free, full control of renditions, posters and AI frames	– We run the workers and queue	Compute only (~$40/mo at growth)	Powers VLC, OBS and most video pipelines
Video (alternative)	Mux or Cloudflare Stream	+ Managed encoding, adaptive streaming and analytics; no workers	– Per-minute pricing; less control over originals	Usage-based (Stream ≈ $5 per 1,000 min stored)	Mux: many video SaaS products
CDN & DNS	Cloudflare	+ Global edge, free TLS and DDoS protection; pairs with R2 and signed URLs	– Advanced rules and analytics need a paid plan	Free plan; Pro $20/mo optional	Millions of websites
Hosting	Render (web + worker + disk)	+ Git-push deploys, HTTPS, disks, background workers; no DevOps team needed	– Fewer knobs than AWS; one region per service	Starter $7 · Standard $25/mo · disk $0.25/GB	Render hosts many startups
Hosting (alternatives)	Fly.io · Railway · AWS	+ Fly/Railway similar; AWS for full control at large scale	– AWS needs DevOps skills	Comparable; AWS varies	—

Not suitable as built: Netlify and Vercel (serverless). They suit static sites and short functions; Glink needs a long-running server for live updates, background jobs and video work.

Money, messages, jobs and visibility.

AREA	CHOICE	WHY IT'S THE BEST FIT	TRADE-OFFS	APPROX. COST	PROVEN AT
Payments	Stripe Checkout + Connect Express	+ Cards, Apple/Google Pay; Stripe runs KYC and bank payouts; built-in idempotency	– Not available to platforms in Nigeria and much of Africa	2.9% + 30¢ per card · Connect $2/active account + 0.25% + 25¢ per payout	Shopify, Lyft, Substack
Payments (Africa)	Paystack (adapter)	+ Cards, bank transfer, USSD; local payouts in NG, GH, KE, ZA; owned by Stripe	– Separate integration; local payouts only	NG: 1.5% + ₦100 per local card (capped)	Thousands of African businesses
Email	Resend	+ Simple API, good deliverability, quick domain setup	– Younger company	Free 3,000/mo · $20/mo for 50,000	Popular with modern SaaS
Email (alternative)	Postmark · SendGrid	+ Postmark: excellent transactional deliverability	– Postmark paid only; SendGrid noisier	~$15–20/mo	SendGrid: very large senders
Jobs & cache	Redis (Upstash/Render) + BullMQ	+ Retries, back-off and schedules for transcoding, email, webhooks; pub/sub for live updates	– Another service to run (managed options exist)	Free tier · ~$10/mo at growth	Redis: GitHub, X (Twitter)
Monitoring	Sentry + Better Stack	+ Errors with release tags; uptime checks and alerts	– Two tools to watch	Free tiers · Sentry Team ~$26/mo	Used by most product teams
CI/CD	GitHub Actions	+ Type-check, tests and build on every push; deploy to staging, then production	– Minutes limits on private repos	Free tier (2,000 min/mo)	Industry default
AI review	Claude (Anthropic API)	+ Strong vision and instruction-following; structured outputs make results machine-checkable	– Per-token cost; optional by design	Per token · measured in pilot	—
Speech-to-text	Deepgram or AssemblyAI	+ Word-level timestamps to check spoken lines without a script	– Third-party audio processing (DPA needed)	Fractions of a cent per audio minute	—
Backups	Disk snapshots → Postgres PITR + R2 versioning	+ Restore to any minute; recover deleted videos	– Restore drills needed	Included / few $/mo	—

Do we need Supabase?

Not for authentication — Glink's own auth already works and is tested; the sign-in problems seen on Render came from the free plan wiping its data on restart. Supabase is a good option as a **managed Postgres** (with a generous free tier) when we move off SQLite. Its Storage is an alternative to R2, but R2's free egress is cheaper for video.

Design principles

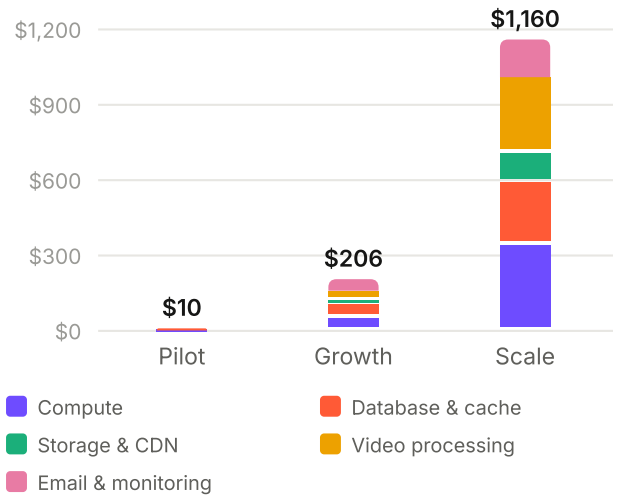
Standard protocols (S3, SQL, OAuth, webhooks) over proprietary APIs; managed services for anything stateful; every vendor behind a small adapter (payments, email, storage, AI) so it can be swapped without touching product code.

4 · RUNNING COST

Pay for what the stage needs, and no more.

Indicative monthly infrastructure cost

USD per month by stage, excluding Stripe fees and AI/speech usage, which scale with revenue



Estimates from public list prices at the time of writing; confirm before budgeting.

ITEM	PILOT	GROWTH	SCALE
Compute	Render Starter $7	2× Standard $50	Web + worker pools ~$340
Database & cache	SQLite + disk $3	Postgres ~$45 + Redis $10	Postgres + replica + Redis ~$250
Storage & CDN	On disk $0	R2 ~1 TB $15	R2 ~8 TB $120
Video processing	In app $0	1 worker $40	Transcode pool ~$300
Email & monitoring	Free tiers $0	Resend $20 + Sentry $26	~$150
Total / month	**~$10**	**~$206**	**~$1,160**

Usage costs on top: card processing (2.9% + 30¢), payouts, AI review tokens and speech minutes — all proportional to transactions, so they scale with revenue.

Scaling stages

STAGE	TRIGGER	ARCHITECTURE
Pilot (≤ 50 brands)	First paying brands	One web service + persistent disk (today), Stripe and Resend on, escrow automation running
Growth (≤ 500 brands)	Uploads over ~150 MB, more than one server, or zero-downtime deploys	R2 video pipeline, Postgres, Redis, one worker, two web instances
Scale (~5,000 brands)	Sustained high upload volume or multiple regions	Autoscaled web and worker pools, read replica, dedicated transcode workers, CDN tuning

Environments and operations

- **Local, staging, production** — staging uses Stripe test keys and seeded data; production is promoted from staging.
- **CI on every push**: type-check, 36+ tests and a build; migrations run automatically after a backup; rollback by redeploying the previous build.
- **Alerts** for errors, uptime, failed webhooks, reconciliation differences and queue backlogs.

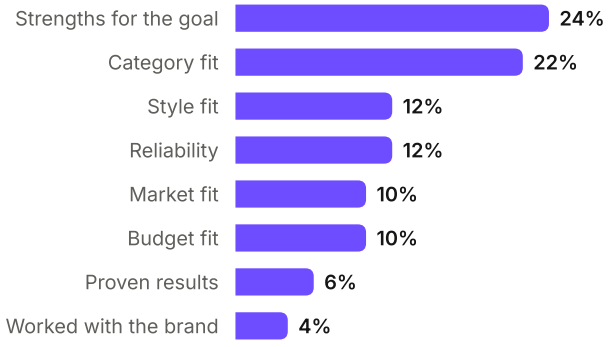
5 · THE MATCHING ENGINE

A score anyone can read, learned from results.

Every creator gets a 0–99 score for every campaign: a weighted sum of eight factors, each producing a plain-language reason. It is a set of pure, unit-tested functions, so weights can be tuned safely as real data arrives.

Factor weights

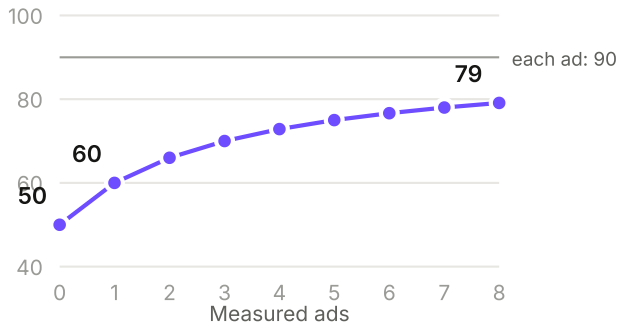
Share of the match score each factor carries



Source: `server/src/matching.ts`. Score = min(99, 100 × Σ weight × factor × availability).

Strengths move towards real results — steadily

Learned hook strength for a creator with a prior of 50 whose ads each score 90



learned = (prior × k + Σ observed) / (k + n), k = 3 (4 for reliability). One lucky video can't swing a score.

Goal decides which strength counts

GOAL	HOOK	RETENTION	CONVERSION
Purchases	25%	25%	50%
Sign-ups	25%	35%	40%
Content testing	40%	40%	20%
Awareness	55%	35%	10%

Who gets invited

- Unavailable creators, or rates above twice the fee, are not eligible.
- Auto-match invites eligible creators scoring **55+**, up to the number wanted; a decline invites the next best.
- Availability multiplies the score: Limited × 0.85; three or more active jobs × 0.8.

How strengths are measured

STRENGTH	FROM
Hook	3-second view rate (≈90 at 50%)
Retention	15-second view rate (≈90 at 35%)
Conversion	Click-through and conversion rate
Reliability	On time (100) or late (40)

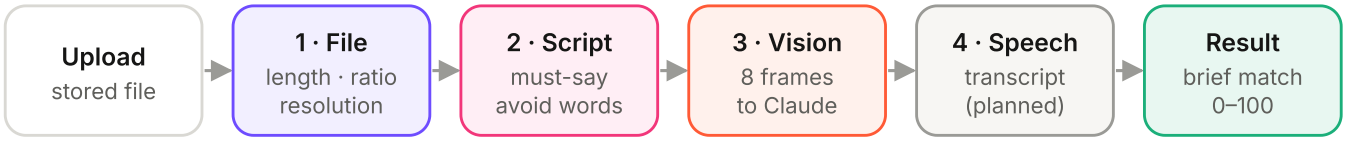
Proposed next

Import results automatically from TikTok and Meta (Phase 4) so strengths update without manual entry; review weights quarterly against which matches produced winning ads; add audience data from verified channels.

6 · PRE-FLIGHT CHECKS AND AI

Catch problems before a brand spends time on them.

Each upload is checked against its brief in layers. Anything that can't be verified is marked "check by eye" — the system never guesses. In the target architecture these checks run in a background worker on the verified file.



pass ✓ · fail ✕ with a fix and timestamp · review ? "check by eye" — the same list is shown to creator and brand

LAYER	CHECKS	STATUS
1 · File	Length within range; aspect ratio within 3% of the brief; resolution (1080p passes, 720p review, lower fails); MP4/MOV	Built moves server-side with ffprobe (Phase 2)
2 · Script	Must-include lines and avoid words (quoted words plus a built-in medical-claims list), with estimated timestamps	Built
3 · Vision	Up to 8 sampled frames judged by Claude for visual items ("show the texture"), returning pass, fail or unclear with a timestamp	Built optional, needs an API key
4 · Speech	Speech-to-text transcript with word timings, so spoken lines are checked without a creator-supplied script; music-over-voice detection	Phase 4 provider such as Deepgram or AssemblyAI

Scoring

Brief match (0–100) = 20 × format score + 80 × brief score, where each item counts pass 1, review 0.6, fail 0.

AI cost control

Frames are down-sampled and capped at eight; one structured-output call per upload; AI can be turned off per environment. Planned: cache results per file so re-checks of the same upload are free. Per-review token cost is measured during the pilot.

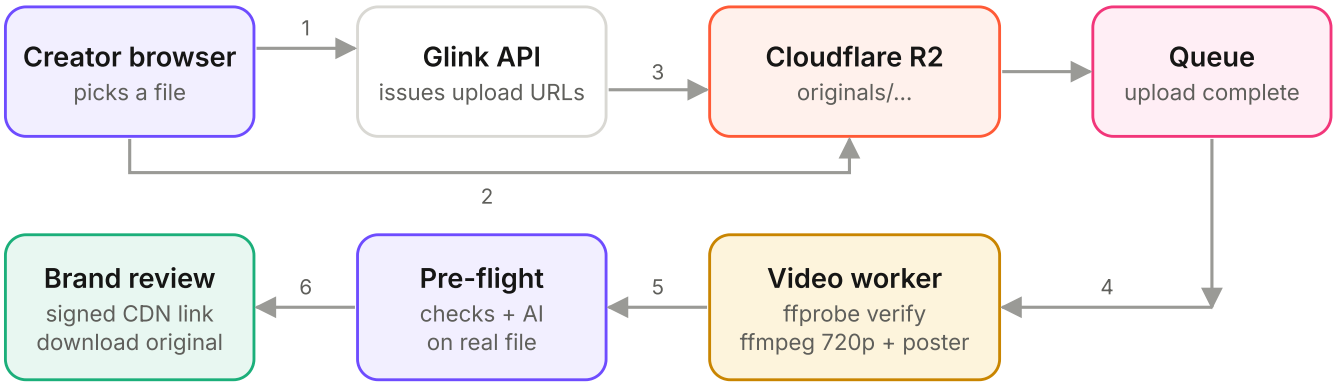
Failure handling

If an AI or transcription call fails, the deterministic results stand and the item is marked for a human — uploads are never blocked by a third party.

7 · VIDEO STORAGE AND DELIVERY

Upload straight to storage. Verify on the server. Stream privately.

Video is the heaviest thing Glink handles and the asset brands pay for. The plan takes large files off the app servers entirely, verifies what was uploaded, produces fast previews, and keeps originals safe for download.



1 request upload URLs · 2 upload 10 MB parts straight to R2 (resumable) · 3 confirm
4 queue a job · 5 verify, transcode, check · 6 notify the brand

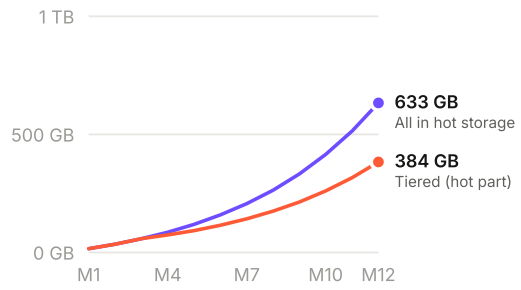
No video bytes pass through the app servers · originals are never altered · playback links expire in minutes

How it works

- **Resumable direct upload.** The API issues short-lived, presigned multipart URLs; the browser uploads 10 MB parts straight to R2 and retries failed parts. Files up to 2 GB; no server memory used.
- **Server-side verification.** A worker runs `ffprobe` on the stored file — real duration, resolution, codec and audio — replacing browser-reported values.
- **Renditions.** `ffmpeg` makes a 720p review copy (fast start), a poster image and the frames for the AI check. The original is never altered.
- **Private playback.** Signed Cloudflare links that expire in minutes; brands download the original on approval.
- **Safety.** File type checked by content, not name; optional moderation scan.

Projected hot storage, first year

Assumes 200 videos in month 1 growing 20%/month (≈7,916 videos), 60 MB originals, 20 MB renditions



Tiered policy keeps renditions plus 3 months of originals hot, and moves older originals (250 GB by month 12) to infrequent-access storage.

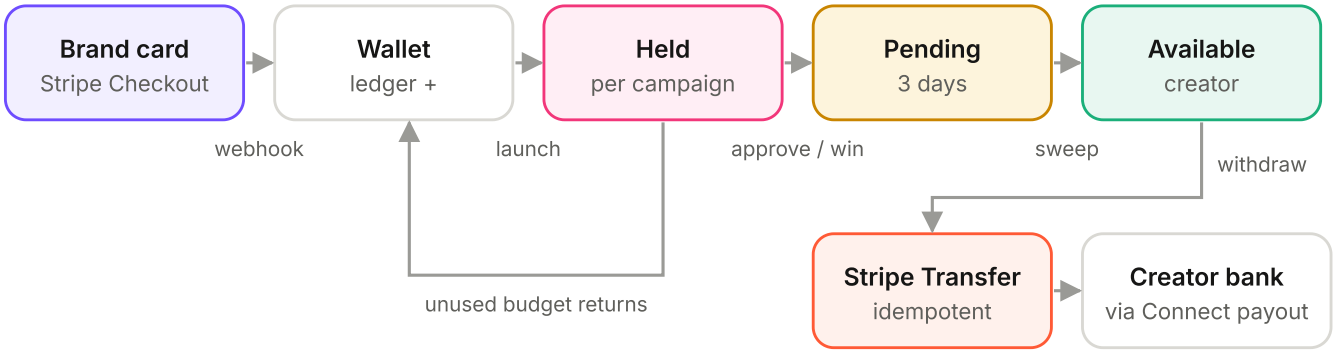
RETENTION TIER	WHAT	RULE
Hot (R2 standard)	Review copies, posters, originals under 90 days	Serves all playback; zero egress cost
Cool (R2 infrequent access)	Originals older than 90 days	Lifecycle rule moves them automatically; restored on demand for re-download
Deleted	Rejected or abandoned uploads; closed accounts	After 30 days, honouring licences and privacy requests

Why R2? Storage at these volumes costs tens of dollars a month; what gets expensive with video is **egress** (every playback), which R2 doesn't charge. Its S3 API keeps AWS S3 + CloudFront open, and a managed service such as Mux remains a per-minute shortcut if we need adaptive streaming later.

8 · PAYMENTS: MONEY IN, MONEY OUT

Stripe moves the money. The ledger explains it.

Brands top up a wallet by card; campaign budgets are held when a campaign launches; fees release on approval and become withdrawable after three days; bonuses and prizes follow the same path. Stripe handles card data, identity checks and bank payouts, so Glink never touches card or bank details.



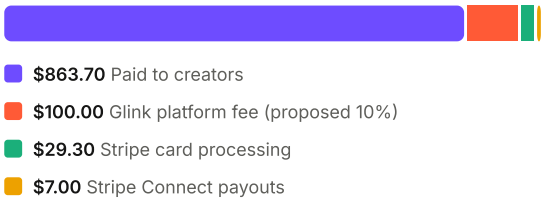
Contest prizes follow approve → pending. Platform fee (proposed) is retained before transfer.

The flow, step by step

EVENT	SYSTEM ACTION
Top-up	Stripe Checkout session (cards, Apple Pay, Google Pay). The wallet is credited only by the signed `checkout.session.completed` webhook, once per session.
Launch	(fee + max bonus) × videos × creators is held from the wallet; launch is refused if funds are short.
Approve	Base fee moves to the creator as pending; available after 3 days (background sweep).
Ad wins	CTR above benchmark within 14 days → bonus paid automatically to pending.
Withdraw	One Stripe Transfer to the creator's connected account (idempotency key = payout ID); Stripe pays their bank.
Complete	Unused budget returns to the wallet; unearned bonuses after their 14-day window.

Where $1,000 of brand spend goes

Illustrative: one US card top-up paid out to two creators in a month



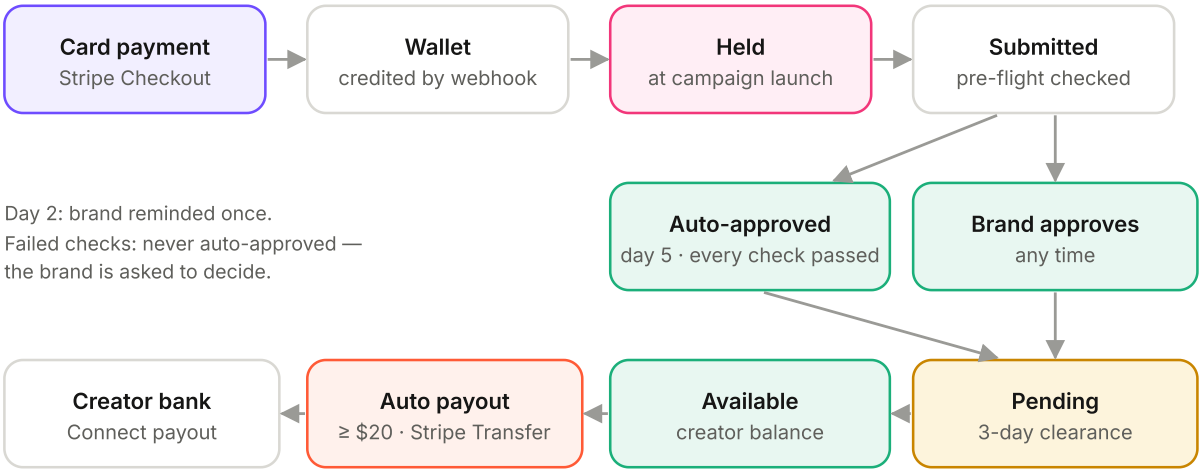
Assumes Stripe US list pricing (2.9% + $0.30 per card charge; Connect $2 per active account per month + 0.25% + $0.25 per payout) and a proposed 10% platform fee. Verify current pricing; cross-border and currency fees vary by country.

Model. "Separate charges and transfers": the platform collects the payment, records it in the ledger, and transfers to creators when they withdraw. This fits Glink's held-budget, release-on-approval flow, and lets Glink take its platform fee before the transfer. Creators onboard to **Stripe Connect Express** — Stripe runs identity verification and stores bank details.

9 · AUTOMATED ESCROW

Money moves on its own — and only when it should.

Nobody chases a payment: a background job runs every minute and moves money forward, while the ledger and idempotency keys make sure nothing is paid twice. Built and covered by automated tests.



Every step is a ledger entry; webhooks and transfers are idempotent, so retries never double-pay.

AUTOMATION	TRIGGER	WHAT HAPPENS	SETTING
Review reminder	Video waiting 2 days	Brand reminded once, told when auto-approval will happen	`REVIEW_REMINDER_DAYS`
Auto-approval	Waiting 5 days, all checks passed	Approved and fee released to the creator; brand notified. Videos with failed checks are never auto-approved — the brand is asked to decide	`AUTO_APPROVE_DAYS`
Clearance	3 days after approval	Earnings move from pending to available (protects against disputes)	—
Bonus	Results beat the brand's rule	Bonus paid automatically — any metric, threshold and window the brand chose	per campaign
Auto-payout	Available ≥ $20, bank connected	Stripe Transfer to the creator's account; creators can switch it off	`AUTO_PAYOUT_MIN`
Return unused	Campaign completes; bonus window closes	Unspent budget and unearned bonuses return to the wallet	—

How others do it

Upwork releases a fixed-price milestone automatically if the client doesn't respond within 14 days. **Fiverr** auto-completes an order 3 days after delivery and clears seller funds after a holding period. Glink uses 5 days to approve and 3 days to clear.

Safe by construction

Webhook events stored once; a checkout credits once; each transfer has an idempotency key; one payout in flight per creator; failed payouts retry on the next run and never lose earnings. At scale the same jobs move to a Redis queue (BullMQ).

10 · PAYMENT CONTROLS AND COMPLIANCE

Correct under retries, auditable, and ready for more markets.

Built today

- Webhooks verified by signature; each event stored once (`stripe_events`).
- A checkout session can credit a wallet only once (unique `external_id`).
- Transfers use idempotency keys; one withdrawal in flight per creator.
- Payout account readiness tracked from `account.updated`.
- Friendly handling when Stripe's available balance is short — earnings stay safe.
- Automated payment and escrow tests run on every change.

Proposed (Phase 1 and 5)

- **Platform fee** configured per plan and shown on every payout statement.
- **Refunds** of unused wallet balance to the original card, recorded in the ledger.
- **Disputes:** `charge.dispute.created` freezes the disputed amount and alerts the owner.
- **Nightly reconciliation** of ledger totals against Stripe balance transactions, with alerts on any difference.
- **Statements and receipts** (PDF) for brands; earnings statements for creators.
- **Audit log** of every money action and who performed it.

Compliance

TOPIC	APPROACH
Card data (PCI DSS)	Card entry happens on Stripe-hosted Checkout, so Glink's scope is the lightest (SAQ A). No card numbers touch Glink servers.
Identity (KYC)	Stripe Connect verifies creators and screens payouts; Glink stores only the connected-account ID and readiness.
Holding funds	Funds sit in the platform's Stripe balance, tracked per brand and campaign. Wording such as "escrow" and holding periods to be confirmed with legal counsel per launch country.
Tax	Stripe's tax forms for creator earnings where applicable (e.g. 1099 in the US); VAT/sales tax on platform fees via Stripe Tax if required.

More markets: a provider adapter

Stripe is not available to platforms based in Nigeria and much of Africa, and payouts to some countries are restricted. We will put Stripe behind a small `PaymentProvider` interface — *create top-up · verify webhook · onboard payee · transfer · refund* — and add **Paystack** (cards, bank transfer and local payouts in Nigeria, Ghana, Kenya and South Africa) as a second implementation. Each brand and creator is routed by country and currency; the ledger records the currency of every entry.

Stripe
US, UK, EU, Canada, Australia and more; cards, wallets; Connect payouts.

Paystack
Nigeria, Ghana, Kenya, South Africa; cards, bank transfer, USSD; transfers to local banks.

Later, if needed
Flutterwave for wider African coverage and mobile-money payouts.

11 · SECURITY AND PRIVACY

Secure by default, with less to protect.

The best protection is not holding sensitive data: card and bank details stay with Stripe, and videos are private by default. What Glink does hold is protected at every layer.

AREA	BUILT TODAY	PROPOSED
Passwords	scrypt with a per-user salt, constant-time comparison, minimum 8 characters	Breached-password check at sign-up
Sessions	Random 256-bit token in an HttpOnly, SameSite=Lax cookie (Secure on HTTPS); only a SHA-256 hash stored; 30-day expiry	Session list with “sign out other devices”
Sign-in	Google, Facebook, Apple; accounts link only through verified emails	Two-factor authentication (TOTP) for brand owners and anyone who can spend
Abuse	Rate limits: 10 log-in attempts per email and address per 15 minutes; limits on sign-up and reset	Shared limits in Redis; bot protection on sign-up (e.g. Turnstile)
Requests	Same-origin check on every change; security headers; product-URL fetcher blocks internal addresses	Strict Content-Security-Policy
Access	Every request checked against the signed-in user and team role	Audit log for money and permission changes
Supply chain	Lockfile, pinned versions, secrets kept out of the repository	Dependabot and secret scanning; signed builds in CI
Assurance	36 automated tests including cross-site, data-isolation and escrow cases	External penetration test before public launch

Privacy and data protection Phase 1 & 5

Designed for GDPR (EU/UK) and Nigeria’s NDPA: a privacy policy and terms at launch; self-serve data export and account deletion; data minimisation; processors (Stripe, Resend, Cloudflare, AI and speech providers) listed with data-processing agreements. Creator videos will not be used to train third-party models.

Content rights Phase 1

Each campaign will record the licence a brand receives (platforms, duration, paid usage, territories), which creators accept before filming. Approval and payment stamp the licence start; expiry reminders help brands stay compliant; paid-partnership disclosure (#ad) becomes a default brief item.

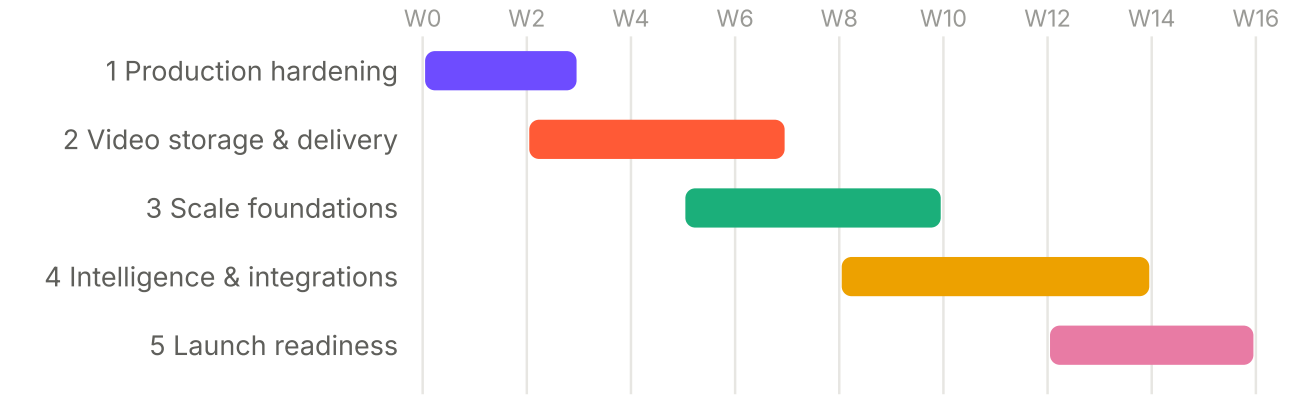
12 · ROADMAP

Sixteen weeks to a production platform at scale.

Phases overlap to keep two engineers busy without blocking each other. Each phase ends in something usable in production, with explicit acceptance criteria.

Delivery plan

Weeks from kick-off; indicative effort for two full-stack engineers plus part-time design and QA



PHASE	SCOPE	DONE WHEN
1 · Production hardening weeks 0–3	Render production with disk (or managed Postgres); Stripe live (top-ups, payouts, platform fee, refunds); Resend; error tracking and uptime; CI; staging; terms and privacy	A real brand tops up, a real creator is paid out, and a restore from backup is rehearsed
2 · Video storage & delivery weeks 2–7	R2 resumable uploads, ffprobe verification, 720p renditions and posters, signed playback, lifecycle rules, migration of existing files	2 GB upload survives a dropped connection; playback starts under 2 s; no video bytes pass through app servers
3 · Scale foundations weeks 5–10	Postgres migration with data copy, Redis pub/sub and shared rate limits, BullMQ workers, two web instances, zero-downtime deploys	Load test at 10× pilot traffic within targets; failover of one instance with no user impact
4 · Intelligence & integrations weeks 8–14	Speech-to-text pre-flight, TikTok and Meta results import, verified channels, insight screens built from real data	Bonuses and strengths update from imported results with no manual entry
5 · Launch readiness weeks 12–16	Penetration test and fixes, two-factor auth, audit log, reconciliation, Paystack adapter if launching in Africa, documentation	Security findings closed; reconciliation clean for 14 days; launch checklist signed off

What could go wrong, and what we need from you.

RISK	LIKELIHOOD	IMPACT	MITIGATION
Payment provider not available in a launch country	High (Africa)	High	Provider adapter; Paystack for Nigeria, Ghana, Kenya, South Africa; decide launch countries early
Stripe balance too low for instant payouts	Medium	Medium	Payouts wait for settled funds; clear creator messaging; optional platform float
Large uploads fail on mobile networks	Medium	High	Resumable multipart uploads; retry per part; size and length guidance in the brief
AI or speech provider outage or cost spike	Medium	Low	Deterministic checks always run; per-environment switches; usage budgets and alerts
Ad-platform API approval delays (TikTok, Meta)	High	Medium	Apply in Phase 1; manual results entry remains as fallback
Data loss on the single-server pilot	Low	High	Disk snapshots + nightly copies off-server; Postgres PITR in Phase 3
Regulatory treatment of held funds	Low–Med	High	Legal review per country before launch; keep funds in Stripe/Paystack balances

Decisions needed

Business

- Launch countries and currencies (drives Stripe vs Paystack).
- Platform fee and plan pricing; who absorbs card fees.
- Default content licence offered to brands.

Technical

- Hosting: stay on Render, or plan an AWS move at scale.
- Postgres provider (Neon, Render, Supabase).
- Speech-to-text provider; enable AI frame checks in production.

Next two weeks

- Production service on Render with a persistent disk, a custom domain and HTTPS.
- Stripe account activation and Connect platform profile; first live top-up and payout.
- Resend domain verification; error tracking and uptime alerts; CI pipeline.
- Apply for TikTok and Meta marketing API access; start the R2 upload spike.

In one line: launch the pilot on what is built, switch on real payments and email, move video to object storage, then scale the data layer when the numbers say so.